

Practice Assignment 2

Q1. Creating and Inspecting NumPy Arrays

Create the following two NumPy arrays:

```
A = [10, 20, 30, 40, 50]

B = [
    [1, 2, 3],
    [4, 5, 6]
]
```

Using NumPy, display the following information for both arrays:

- The array itself.
- Number of dimensions using `ndim`.
- Shape using `shape`.
- Total number of elements using `size`.
- Data type using `dtype`.

Q2. NumPy Array-Creation Functions

Create an array containing 20 elements using each of the following NumPy functions:

- `np.arange()`
- `np.linspace()`
- `np.zeros()`
- `np.ones()`

For each function:

- Display the resulting array.
- Specify the shape of the resulting array.
- Explain what the function does.
- Give one practical application where the function could be useful.

Q3. NumPy Indexing and Slicing

Create the following matrix:

```
A = np.arange(1, 26).reshape(5, 5)
```

The resulting matrix will contain values from 1 to 25. Using only NumPy indexing, slicing, and Boolean masking, perform the following operations:

- Extract the first row.
- Extract the last row.
- Extract the first column.
- Extract the last column.
- Extract the middle 3×3 matrix.
- Extract all even numbers.
- Extract all values greater than 15.
- Reverse the order of the rows.
- Reverse the order of the columns.

Q4. NumPy Data Types and Type Conversion

Create the following NumPy array:

```
A = np.array([10, 20, 30, 40, 50])
```

Convert the array sequentially into the following data types:

- int32
- float64
- int16

For each conversion:

- Display the array.
- Display its data type.
- Explain why data-type conversion may be useful in numerical computing.

Q5. Array Reshaping

Create:

```
A = np.arange(1, 25)
```

Reshape this array into each of the following shapes:

- 2×12
- 3×8
- 4×6
- 6×4

For every resulting array:

- Display the matrix.
- Display its shape.
- Display its size.

Q6. Vectorized Student Marks Processing

Consider the following student marks:

```
marks = np.array([
    45, 67, 89, 32,
    76, 91, 55, 43
])
```

The university has decided to provide 5 bonus marks to every student. Using NumPy:

- Add 5 marks to every student.
- Ensure that no student's marks exceed 100.
- Calculate the average marks.
- Find the highest marks.
- Find the lowest marks.
- Find the number of students who scored at least 50 after adding the bonus.

Q7. Boolean Masking and Conditional Data Modification

Consider the following temperature dataset:

```
temperature = np.array([
    32, 28, 35, 40, 25,
    30, 42, 27, 38
])
```

Using NumPy Boolean masking:

- Extract all temperatures greater than 35.
- Extract temperatures between 28 and 38 inclusive.
- Count how many temperatures are greater than 35.
- Count how many temperatures are below 30.
- Replace every temperature greater than 40 with 40.
- Replace every temperature below 28 with 28.
- Display the final modified array.

Q8. Statistical Analysis Using NumPy

Given:

```
data = np.array([
    12, 15, 18, 21,
    25, 30, 35, 40
])
```

Using NumPy, calculate:

- Mean
- Median
- Variance
- Standard deviation
- Minimum
- Maximum
- Range
- 25th percentile
- 50th percentile
- 75th percentile

Q9. Broadcasting in NumPy

Consider:

```
A = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])

B = np.array([1, 2, 3])
```

Using NumPy broadcasting:

- Task 1: Add B to every row of A.
- Task 2: Multiply every row of A by B.
- Task 3: Subtract B from every row of A.
- Task 4: Divide every row of A by B.

Finally, explain:

- What is the shape of A?
- What is the shape of B?
- Why does broadcasting work here?
- What would happen if B had shape (2,) instead?

Q10. Vectorized Grade Assignment

Given the following marks:

```
marks = np.array([
    35, 78, 92, 41,
    56, 29, 88, 67
])
```

Assign grades according to the following rules:

Marks Range	Grade
90–100	A+
80–89	A
70–79	B
50–69	C
35–49	D
Below 35	F

Using NumPy:

- Create a grade array.
- Display each student's marks and corresponding grade.
- Count the number of students in each grade category.